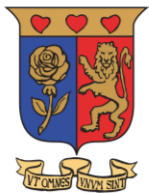


Basics of Electronics for IoT

ICS 4111: Topic 3

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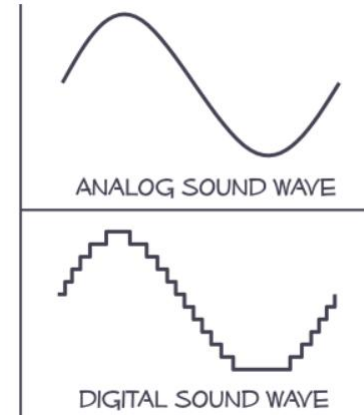
Learning Objectives

- Apply fundamental concepts in electronics:
 - Electronic signals
 - Laws in electronics: **Ohm's law** and **Kirchhoff's law**.
- Identify basic components of electric circuits: **resistors** and **capacitors**, **voltage divider** and the **current limiter**.



Electronic Signals

- **Signal:** function that conveys information about a phenomena. Can be represented in:
 - *analog* format would match the original signal.
 - *digital* format which creates a sequence of steps.
- The representation of a signal is a stream of measurements. The precision of the measurements depends on two important factors:
 - *number of bits* in the measurements.
 - *sampling rate* of the measurements.



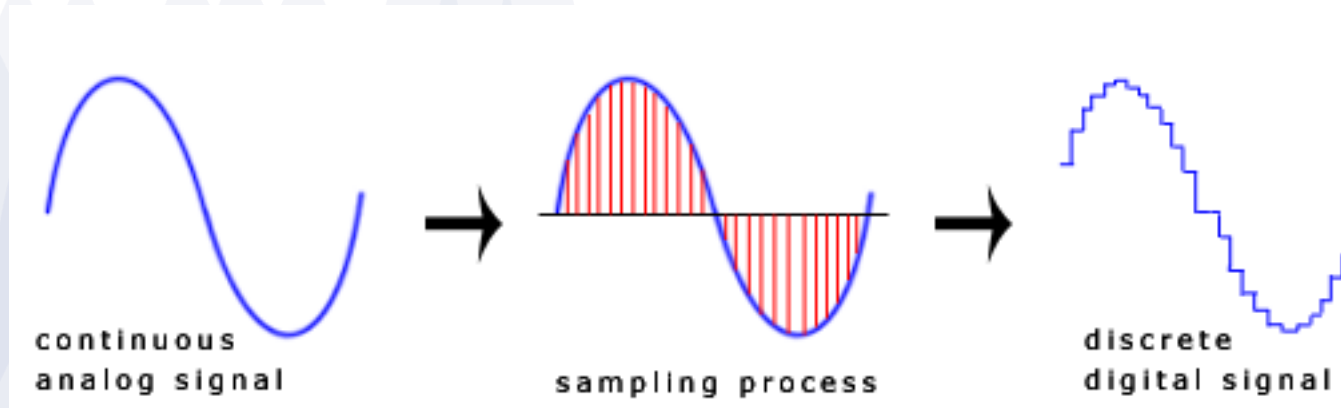


Electronic Signals: Number of Bits

- The *number of bits* assigned to a measurement will either improve or damage the quality of the received signal.
 - The signal is then recomposed after gathering multiple samples.
- In hardware, each sample/measurement, is allocated one or more bits:
 - A one-bit sample will split the signal's values into two parts: above half and under half: Above half gets transformed to 1 and below half to 0.
 - Unfortunately, this signal's representation will be rather inaccurate but if the hardware can have a larger scale of bits, the hardware can adapt the signal closer to the original one

Electronic Signals : Number of Bits

Supposing your hardware allocates n bits to represent your signal: when taking a sample, the hardware decides which of the values from 0 to 2^{n-1} to assign to the measurement.



Electronic Signals : Sampling Rate



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- The ***sampling rate*** is derived from the number of samples of a signal registered during an interval of time.
- During this interval between measurements, the signal is lost, causing a need for approximation.
- The ideal case is when the sampling is **done frequently**, so little signal is missed.
- The **Nyquist theorem** says that:
 - *to reproduce an accurate signal, you need to sample at least twice as fast as the highest frequency of the signal.* This means that if the bandwidth is lower than the highest frequency, the signal will be altered.

Laws of Electricity:

Ohm's Law

In a circuit, the current (I) is directly proportional to the applied voltage (V) and inversely proportional to the resistance (R) of the circuit.

$$V = I * R$$



Laws of Electricity:

Kirchhoff's Law



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Kirchhoff's First Law: Sum of current entering a node (positive) and leaving a node (negative) is 0. I_1 & I_2 is entering while I_3 is leaving

$$I_1 + I_2 - I_3 = 0$$

Note:

currents have directions. Currents incoming have negative values, while currents outgoing have positive values.

Laws of Electricity:

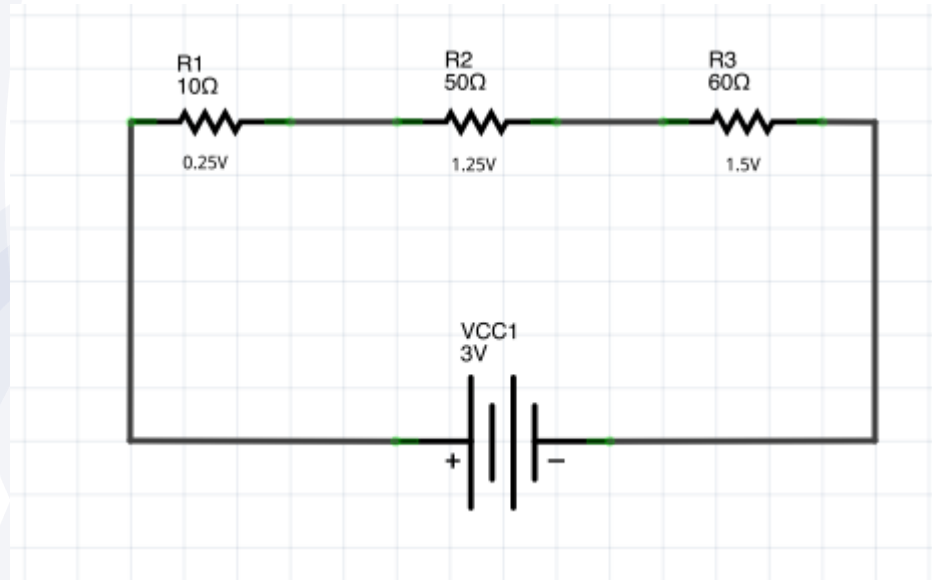
Kirchhoff's Law

Kirchhoff's Second Law states that the sum of the voltage in a circuit loop is equal to the power source voltage/zero. The assumption is that voltage drops across each passive element in the direction of current flow

$$\sum R_k I_k = VCC$$



Kirchhoff's Voltage Law



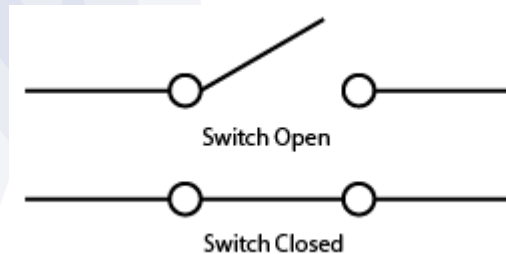
You have a 3V source and three resistors of different resistance. The sum of voltage drops on each of them is equal to the source voltage.

$$R1 + R2 + R3 = VCC1 \Rightarrow 0.25v + 1.25v + 1.5v = 3v$$



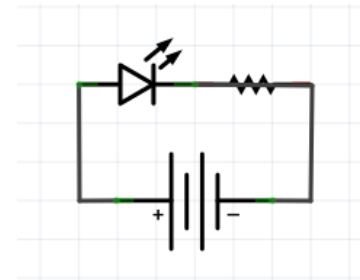
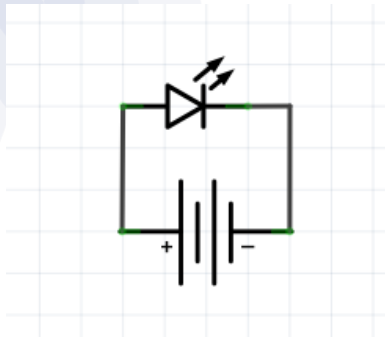
Buttons/Switches

- An electric component that can break an electrical circuit by interrupting the current.
- Symbols for switches in schematic diagrams:



Light Emitting Diode (LED)

- This is an electronic component that has a positive and a negative side, allowing current to flow only in one direction, from positive to negative.
- The LED is also a diode. When current is flowing through the LED, it lights up.
 - Therefore, to light up a LED, you need to put the high voltage at the anode and the low voltage at the cathode.



That's Part 1!

Any Questions?



Scan this code as you take a quick break



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Part 2: Components in Electric Circuits!



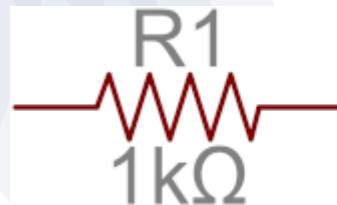
Resistors

- Electronic components that limit the flow of electrons through a circuit.
- They are passive components, meaning they only consume power and can't generate it.
- Resistors are usually added to circuits where they complement active components like op-amps, microcontrollers, and other integrated circuits.



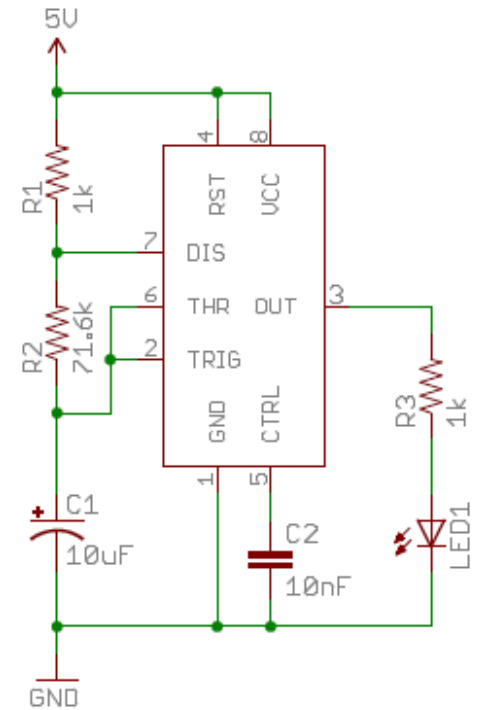
Resistors

- Commonly, resistors are used to limit current, divide voltages, and pull up I/O lines.
- All resistors have two terminals, one connection on each end of the resistor.
- On a schematic diagram, a resistor will show up as one of these two symbols:



Resistors

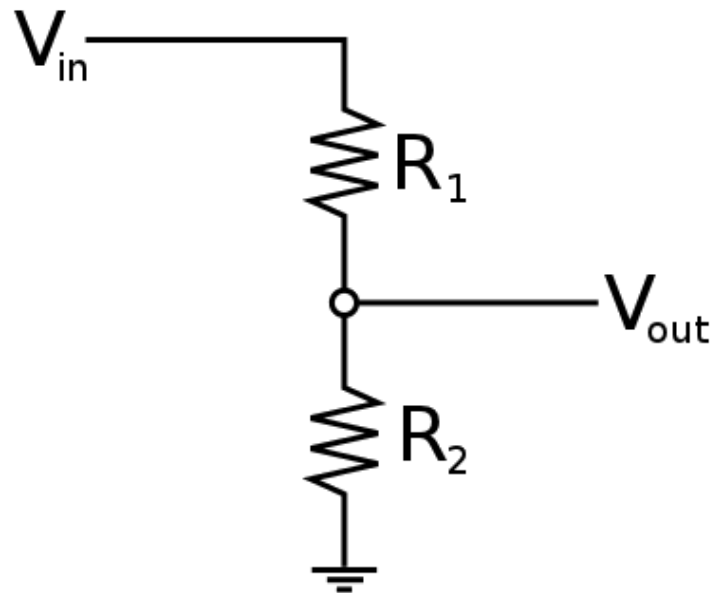
- The name of the resistor is usually an R preceding a number. Each resistor in a circuit should have a unique name or number.
- In this circuit, resistors play a key role in setting the frequency of the 555 timer's output.
- Another resistor (R_3) limits the current through an LED.





Voltage Divider

- A voltage divider is a circuit that turns a large voltage into a smaller one using two resistors and an input voltage. This way, we can obtain an output voltage.





Resistors & Output Voltage

- One of the resistors should always differ from zero; otherwise, we will have a short circuit. The output voltage can be calculated using the following equation:

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} \cdot V_{\text{in}}$$

Find an online tool for voltage calculation [here](#).

Capacitors

- A capacitor is a two-terminal electrical component.
- Along with resistors and inductors, they are one of the most fundamental passive components we use.





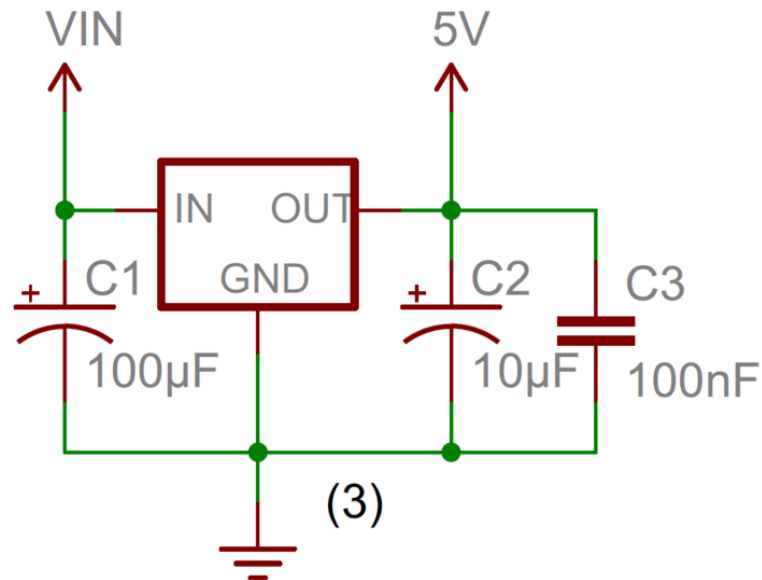
Capacitors

- Capacitors specialty is their ability to store energy; they're like a fully charged electric battery.
- Common applications include local energy storage, voltage spike suppression, and complex signal filtering.



Capacitors

- The capacitor symbol consists of two parallel lines, which are either flat or curved; both lines should be parallel to each other, close, but not touching (this is representative of how the capacitor is made).





Capacitors

- Each capacitor is built to have a specific amount of capacitance (amount of charge it can store)
- The standard unit of capacitance is called the farad, which is abbreviated F.
- 1 farad is a lot of capacitance; even 0.001F (1 milifarad = 1 mF) is a big capacitor.
 - Usually, you'll see capacitors rated in the pico (10^{-12}) to microfarad (10^{-6}) range.



Capacitors

- Much like resistors, multiple capacitors can be combined in series or parallel to create a combined equivalent capacitance.
- Capacitors, however, add together in a way that's completely the opposite of resistors.
- When capacitors are placed in parallel with one another, the total capacitance is simply the sum of all capacitances. This is analogous to the way resistors add when in series.

Calculation of Resistance



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Resistors in series

$$R_1 + R_2 + R_3 = R_{total}$$

Resistors in parallel

$$\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{R_{total}}$$

Calculation of Capacitance



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Capacitors in parallel

$$C_1 + C_2 + C_3 = C_{total}$$

Capacitors in series

$$\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{C_{total}}$$

That's all for now! Any Questions?

Next – Topic 4: Microprocessors & Microcontrollers